Super Solar Storms with Lika Guhathakurta

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis This episode of Star Talk features astrophysicist Lika Guhatakuta discussing heliophysics and the Sun as a living laboratory, addressing questions regarding mass loss rates through fusion and solar wind, acoustic properties within stellar interiors despite space's vacuum, and the extent of Earth’s magnetic protection via the heliosphere. Through analysis of recent missions like Parker Solar Probe and data from global observatories, the episode explores how 11-year solar cycles drive auroral phenomena on a planetary scale while clarifying misconceptions about mass loss gravity reach, cosmic radiation shielding, and international scientific collaboration in space weather monitoring.

2. Key Arguments & Data

3. **Heliosismology:** Solar seismology is treated as "ultrasound" for the Sun; scientists infer interior structures from vibrations (seismic waves) that bounce off boundaries within plasma rather than solid rock because 99.9% continuous observation data via SDO and ground stations in New Mexico prevents erroneous results.
4. **Mass Loss Metrics:** The Sun loses mass at a rate of roughly 5 to 6 million metric tons per second; this includes ~4M tons from fusion (hydrogen-to-helium) plus ~1–2M tons carried by the solar wind. Over its lifetime, total loss remains an order of magnitude below 0.03% of original mass, maintaining structural integrity despite continuous leakage.
5. **Solar Acoustics:** The Sun generates overlapping pressure waves/oscillations with a frequency of approximately five minutes (five-minute oscillation), which Neil deGrasse Tyson describes as sounding like a constant "loud roaring lawnmower" if sound could propagate through the vacuum to Earth, though space acts as natural insulation.
6. **Heliosphere Geometry:** The heliosphere is not spherical but asymmetric in shape and size due to interactions between solar wind pressure/magnetic fields and interstellar medium; Voyager spacecraft

have reached this boundary (heliopause), which functions as a protective cocoon for the planetary system against galactic radiation.

7. **Magnetosphere vs Heliopause:** Boundaries like magnetopauses generally reflect physical laws where gravity, pressure, or magnetic sweeping define limits while Earth's gravitational pull extends forever into space relative to its mass dominance over local objects (e.g., Moon).
8. **Solar Cycle Dynamics:** The Sun follows an 11-year cycle for sunspot intensity and a 22-year full polarity flip; during solar maximum/G4-G5 storm levels, the aurora becomes visible at latitudes far beyond magnetic poles due to energy interaction with ionosphere (May/November 2024 events).

9. **Space Weather Risks:** A Carrington-level event would not necessarily destroy infrastructure if satellites are properly hardened or shut down; however, CME-driven drag can affect launch orbits for Starlink arrays as seen in late January/February cases where ~39 of 49 launched satellites failed.

10. Actionable Takeaways

11. **Monitor Solar Activity:** Watch NOAA geomagnetic storm scales and look for aurora visibility during G5-level solar maxima events, especially if living at moderate to southern latitudes near magnetic axes rather than geographic poles (e.g., Long Island/DC experiences in May 2024).
12. **Follow Mission Data:** Track updates from Parker Solar Probe (currently active, measuring pristine corona wind), IMAP (Interstellar Mapping and Acceleration Probe) operating at L1, POLARIS constellation satellites for coronal imaging gaps, and ESA/NASA joint missions like Solar Orbiter to understand heliosphere boundaries.
13. **Engage in Space Awareness:** Use planetary magnetic axes as guides rather than geographic ones when observing cosmic phenomena; utilize Earth's atmosphere only as a buffer against extreme solar radiation unless you are equipped with shielding.

Full Transcript

So, the sun is still at it. But what's it really up to, Neil? Lika Guhatakurta knows. Yes. Coming up on Star Talk. Welcome to Star Talk. Your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk. Neil deGrasse, Tyson, your personal astrophysicist. Got Chuck nice with me. Hey, baby. That's right. Your personal ass. Astrophysicist. Deal with you later on that one. So, Chuck, we have a returning guest. Yes, someone we very much like. One of the all-time favorites. Yes, a start-talk favorite. Yeah, we have Lika Guhatakuta. Yes. I even think I pronounced it the right way. You did it, right? You did it. Lika. Welcome back to Star Talk. Thank you. And it's great to be here and great to have Chuck in the show. What can I say? Well, thank you. You're one of the world's experts on the sun. And those who are consuming this via video, you may notice that her hair is as bright as the sun. Yes. Yes. Sun kissed, as they might say. Sun kissed, perhaps. Yeah. Don't go there.

So you're an astrophysicist and the technical name for your subspecialty I guess would be Helio physicist. Do we got that right? You got that right. All right. A brand new word about 20 years old. Oh, cool. Oh, I did not know that. Did not know that either. Brand new discipline word perhaps people have used it. But it's a brand new discipline. A thing unto itself. There you go. Except you were interested in the sun long before that. So what attracted you? Just up because we like a little bit of bio on people every now and then and someone with such intense interest in such an omnipresent object. You can ask why isn't everybody interested in it or how come more people aren't? So what got you into it?

You know, it's not a traditional approach, right? So I'll give the most honest answer I can give, which is, I came to the sun almost by accident. I was trained as an astrophysicist, interested in stars in general, and you know what that is all about, Neil. Then I realized that we spend studying stars that are light years away and you know we have one star which sits only eight light minutes from us and allows us to examine it in exquisite detail so the sun became the most exciting laboratory i could imagine we can see the surface change listen to its interior fly through its atmosphere of late and

watch its influence literally move all the way to Earth and the edge of the solar system, which is our Milky Way galaxy. Okay, so right now, do you live in the sun? Is this your... And you just come out for interviews. We do live in the atmosphere of a living, breathing star.

What did they say? Look at that. See what she did there. That was good. So now, I remember coming through graduate school, there was a subfield of heliophysics called heliocysmology. And they were, you know, barring the word from earth, you know, seismology and earth. As you watch earthquake waves move through the earth and they refract and bend and you can infer what the middle of the Earth is like. Okay. From the delayed signals from different points on Earth's surface. And so that made sense to me because Earth is solid. The Sun is not solid. So when I heard them talk about this, I was like, yeah, okay, let's see what you got. But it's a real subfield. Could you tell me how that works? And is what's going on in the middle of the Sun settled science at this point? Ooh. That's well, you have to give me some time.

A lot of questions kind of coupled together. Heliosismology is very much a well-established subgroup of heliophysics, solar physics. It's exactly how you describe. Now, sun is not solid. I think we all know that. But there are vibrations, sound vibrations taking place in the interior of the sun. And that's what we... image the vibrations, and then infer essentially the interior properties of the sun, like the convection zone. You can't see inside the sun. Everything has to be inferred. That's, inside the sun, everything. So is our all scientists who study this inferring the same thing so that there's general agreement about what the middle of the Sun is up to? Or is it like dentists where you have four out of five dentists surveyed says that you should brush your teeth. I don't know who that fifth dentist is. Let me say that we do our best. We are doing ultrasound on the Sun. Okay. It's not a dental office. That's what we are doing. So seismology is like really ultrasound and the more observations you have, more continuous observations, that's why Solar Dynamics Observatory was created. We needed 99.9% of the data continuously because any data gap can lead to erroneous results. So SDO, the Solar Dynamics Observatory, many of the images we've seen of the Sun's surface in high detail are credited to that observatory. I assume that's in space somewhere, correct? It is in space somewhere. It's actually near our

space because the volume of data is so high, we have our own dedicated ground station in New Mexico. So we couldn't send it to like L1, but we are watching the sun nearly 24-7. There are eclipse periods because of its orbit being close to Earth, but yes.

There were a lot of thought. So if it was much, much closer to Earth, it would have daytime and nighttime, but you want near 24-7 observations. Obviously, the farther away from you, the less Earth is going to get in the way. Right. Or anything else. Yeah. Well, it turns out in this episode, we were mining the questions that had accumulated in our cosmic queries search, and some of them are Sun related and we exhume them for just your arrival here, and then we'll get on with my conversation with you. So, Chuck, you got questions? I got, we just got a few that are specifically for Lika. These are from our Patreon supporters. From our Patreon patrons, and by the way, whenever we get the opportunity to say thank you to the Patreon patrons, we definitely offer our heartfelt gratitude because, you know, you give us the money, the money. Don't be so crass. Just say you give us support. I'm sorry.

you freely give of yourself to support us. You give us your ideas. That's what they're doing. That's what they're doing. That's true. They do because they offer their questions and perspectives. Okay, this is Lothar Erkins. I love the name Lothar, that's so cool. It's like a caveman name. Yeah, yeah, of course. Lothar. Lothar. Hello, Dr. T, Lord Nice and Leica. Lothar from Edwardsville, Illinois. Has anybody ever crunched the numbers on how much mass our son has lost since igniting the fusion process and recorded the results? Just curious. That's weird. Now, but the real question is, because the sun is radiating, is it indeed losing mass as it's radiating? That's the whole shebang. That's the shebang right there. Yeah, so the real question for Leica is, is this something you guys care about, because you can calculate this. Right. Very simply actually. Okay. We know how much energy is coming off the sun every second. Okay. Okay. Convert that into mass. Mass.

Because it's making that energy by equals MC squared right and now you that's how much mass is losing per second They ask how many seconds has the Sun been alive right? It was a little brighter in the past but and then you

get a total mass So Leica is this something that matters to the heliophysicist the rate at which the Sun is losing mass? Because I think the rate is pretty slow. Well, it's not so much the rate But the fact that it is losing mass how much mass in what ways? So the Sun has two main leaks, you already mentioned one. Fusion of it converts 600 million tons of hydrogen into helium. Of that, 4 million metric tons of mass is converted into energy. That's every second, you know, it's equal to mc squared that you mentioned. That's what's going on.

That's amazing. But there is another one that people might not know about and that is that solar wind carries away roughly another 1 to 2 million tons of matter each second. So altogether the sun at this present time is losing roughly between 5 and 6 million metric tons basically, right? Per second. sounds alarming until you actually remember the size of the sun its mass and over its lifetime you know people have calculated because all these scientists are basically known so you can imagine someone calculated this somewhere and it's not a difficult calculation. So over its lifetime the total is only sort of an order of 0.03% of the total mass of the sun. So sun maintains its weight very well. Yeah. Unlike me. Sun knows so. You're a creating mass. I am a creating mass while the sun is maintaining its mass. Okay, so if the sun has strong gravity, as we all know and love it, that's what keeps us in orbit. Yeah. How is it losing particles? Why don't the particles just stay with the sun? What accelerates them and sends them out? You are.

jumping to all kinds of simple science questions which I haven't even gone to. So you are absolutely right. I mean, everything, you know, on the sun at all things are governed by basically gravity, pressure, magnetic field close to the sun that we call the corona that the particles are dominated by gravity and magnetic field. And then we get to a point where the temperature in the corona rises so much, pressure increases, and particles become what we call the solar wind. And that region is called the alphanic surface or alphan critical point, where the particles are no longer tethered to the sun by gravity. and I support Star Talk on Patreon. You're listening to Star Talk with Neil deGrasse Tyson. All right, this is Corey Haas, who says, hey, simple question. Does the sun make noise if space allowed for sound to travel? Well, you already mentioned that we do some ultrasound measurements. So if sound can move through it, I think this question is,

Anything we can listen in on, is there some, or if there was some way for sound to cross the vacuum of space, what would the sun sound like? It's a deep, deep question, Neil. I find all sun questions to be deep, by the way. I love these questions. They seem ordinary on the face of it, but really you have to think about it, right? So the answer is the sun is not.

quiet I mean imagine as if I've said something it rings like an enormous musical instrument with millions of overlapping pressure waves that I talked about that creates the field of phyliosysmology and a strong family of these sort of oscillations around five minutes. And five minute oscillation is kind of has been known for a while when we started actually measuring these surface waves. Now, Neil, you already said that we cannot hear that sound directly because the vacuum between the sun and earth does not carry ordinary acoustic waves. And I would say, thank God for that. Space is nature's sort of finest soundproofing, cancelling device outside of the Bose acoustics, right? Yeah. I wish my bedroom was space. We still listen by, as I mentioned, by measuring the tiny motions and we do it with ultrasound. So if, if, and this is, this is like, you know, I can't be certain about it. But if we could answer your question, you'll somehow get those sound to propagate in our environment, it is thought that that would sound like a loud, roaring lawnmower. Lawnmower? Yes. Only on weekends. Don't do that during the weekday. Or at least it's not a leaf blower. It would be constant, okay? It's not going to stop. That's the other thing, right? So be grateful, Chuck.

that space is a vacuum. Yes. Otherwise, we'd be living with that. Now, see, I would have thought it would have sound like a frying egg. Oh, quick, quick. Like, that makes more sense. But a lawn mower, that's crazy. So some of the big waves will do that. Either little ones could be crackling like an egg, you know. So a couple of things. I just want to re-emphasize the concept of a pressure wave. Yeah, I was about to. So when I speak, It creates a pulse of energy in the air that is a pulse that moves forward and backwards. And so it's that forward and backwards pulse that hits your eardrum to vibrate the eardrum. That causes it to vibrate to the eardrum. Right, and that's how you move sounds. So now normally when you think of a pressure wave, not just from loud speaking, when you hear pressure wave, often that is associated with an explosion.

Okay, yeah, so There's an explosion and emanating from the explosion is the pressure wave which is what does all the damage right exactly And so extreme pressure way extreme pressure wave. Yeah, so this is why if it sound is too loud It'll blow out your eardrum right the pressure wave is too high right on right, but so but here's what I want to know Is the sun with these pressure waves, is that the result of like explosions on the sun? What is the cause of these pressure waves? Do you analogize it to a musical instrument? Who's playing? Where's the orchestra? Well, I mean, the deep-seated orchestra is always at the core, but I'm not going there. It's the convection zone, where there is like a pot of boiling plasma. Okay, that creates the magnetic field.

and everything outward that we experience. So these five-minute oscillations that you describe, that means something in the sun is repeating every five minutes. Is it one of these sort of convective cells or something else? Yes, if you pattern, if you actually measure this, right? I said there are many resonant... Yes, overlapping. So you see this. Wow, okay. That is wild. Okay, and let me ask you this. One of the images I saw from the SDO, it was so close up, and you could see the surface of the sun as though it was a gurgling, boiling, it was like a cauldron or something. Nice. And this, it's almost mesmerizing. You're just watching it boil. And of course, if you were on the sun, you might hear it, but you would vaporize first. You're going to fall down. There's no surface. Oh yeah, yeah. You'll just descend. It's got strong gravity. I mean, so many ways to think about this, right? Strange things will happen. Interesting. That is interesting. Chuck, I think we have one more question before we resume the episode.

As intended, what do you have? All right, here we go. This one is from Monopoly World. That's their handle. That's the handle that's given. All right. And Monopoly World says this, please tell me everything about the heliosphere. Yeah, do you have five hours? Really? I thought nobody would ever ask that question. It's about time. Seriously. I love this question, actually. And yeah, five hours would be an adequate amount of time. Nily, you want to introduce the subject, and I'll follow. Let me stoke the engine, and then she can drive the train where it needs to go. All right, so we'll crank it up. Allow me to remind people that the Greek word for the sun is

helios. Helios. OK? And it's where we get the name for the second element on the periodic table. Helium.

discovered on the Sun before it was discovered on Earth. Oh, wow. So it was of the Sun, Helios. And so Heliosphere, you can ask, there's the Sun, but how about everything outside the Sun? Does the Sun have any influence there? And if it does, let's still connect it back to the Sun. So, Leica, correct me if I'm wrong. Everything that the Sun touches, I feel like, or Lion King. Wherever the sun touches is your kingdom. Right. Okay. Wherever the sun influences particles and magnetic fields beyond itself is the heliosphere. Is the heliosphere. I think that's right. Is that correct? It is correct. Okay. It is correct. That's our entire solar system, isn't it? Well, let's find out. And the boundary too. And the boundary as well. Yeah, it's very interesting. Remind me to give this analogy.

but you know simply put we all know that I think we all know that earth has a magnetic cocoon or a bubble and think of it either way we can see it touch it but we feel it magnetically and it's called the magnetosphere well the sun creates a much larger cocoon or bubble with the solar wind solar wind doesn't stop anywhere It continues to propagate, right? And it's a bubble that surrounds all the planets and presses against the local interstellar medium. This is like the boundary with our own Milky Way galaxy. That is the heliosphere. And I'd like to say that we have actually punched through the heliosphere. That is, I mean, NASA spacecraft launch in the 70s have done that, right? So in the comedy worlds, they say, you are punching up. We are punching out. Punching out. Yes. So they are in the interstellar medium. And this boundary, just like we learned, that the Alpen surface boundary, we learned with Parker Solar Probe.

that we used to think of as a spherical surface. That's what we scientists do. We start with the simplest assumptions till observations tell us something else. What is the same thing here? This, because the two voyages spacecraft went in two different directions, latitude, we know that the heliosphere is not a symmetric boundary, basically. So it's a little closer on one side than on another side. And they respond to the solar wind pressure, magnetic field. And this is, heliosphere is really our home. It protects us from the harmful cosmic radiation from the interstellar medium. So technically the

heliosphere is the solar system. Don't be so planet, planet biased about it. Yeah. Thank you. Well, you know, you know, it is it's a star's environment. I mean, other stars have environment bubbles. They are cholesterol spheres. So this is suns. OK, so what about the magneto pause? Is that the sun going through like a change of life? We pause, we pause. I'm going to magneto pause.

Oh, magnetopause is killing me. You pause there, you get your breath, then you keep moving on. But these boundaries tend to have the word pause in them if I remember my vocabulary. Yes, they do. And so basically what some magnetopause is kind of used in reference to Earth's magnetosphere, heliopause is in reference to... suns, magnetosphere, which is the heliosphere, et cetera. But what they're trying to do, they essentially describe physics laws, right? So, you know, is it gravity dominated? Is it pressure dominated? Is it magnetic fields sweeping out? Those are the conditions that these boundaries define. I remember the day it was reported when the Voyager 1, I think, exited the heliosphere. Right. So it's the heliopause at the end. And we were all excited, but we asked, well, how does it know? And so it has these sensors on board that knows where the sun is because of the sun's influence on them. But then you reach a point where the sensors went haywire. Right. And couldn't find the sun. Couldn't find the sun. They didn't know which way the sun was because...

The sun's influence was dropping while the galactic influence was rising. There's a point where they were about the same. You could no longer uniquely point which way the sun was. We said it has left the solar system. It's like a compass needle behaves, right? If you take it to the North Pole, what will it show? Which ways are gonna point yeah, well, I think it just points every way Yeah, if you if you to personify that oh my gosh I'm without direction in my life So when you look at the heliosphere Okay, and then you get to the the pause and now you have the Galactic influence, right? Okay. Now, is the gravity of our star still influencing anything beyond that point as well? Yeah, that's a simple thing. The gravity goes forever. Goes forever. Forever. Chuck, you're so smart. My God. Nobody bring dare bring up such questions. Well, thank you. Awesome.

So the sun's gravity extends forever, but so does Earth's gravity. So does the moon's gravity. So all that really matters to you is that somebody else's gravity that's dominating where you are. And we exploit that when we go to the moon. We only give the spacecraft enough energy to get to the point where the moon's gravity takes over, and then it falls towards the moon. You don't need anything more than that. I don't need rocket fuel to fall towards the moon. So at that point, of course, Earth's gravity is still there, but it's dominated by the moon's gravity when you're close enough to the moon. So that's the gravity thing. So our sun's gravity, now here's it, is there something that could fall into the influence of our sun's gravity so far out that it pulls it into our solar system?

Oh, yeah. Oh, yeah. Oh, yeah. Well, that isn't that subject for another topic. That's another show. That's another show. A lot of controversy attached. But it's kind of fun. It's really fun because we are beginning to look everywhere. And what we had taken for subtle signs is very unsettled. Love that. And unsettling. And that too, that too. But may I add, you know, heliosphere is my favorite thing, right? May I add a couple of more? Please, please do it. Because you were instructed to say everything there is to know about the heliosphere. That was your instructions. So please proceed. So heliosphere literally partly shields the planets from galactic cosmic radiation. And so before you reach interstellar space you first have to leave home and that is our home and and two thoughts as I was thinking about this topic when I said bubble magnet sphere is a bubble or a cocoon and same thing with Sun's magnetosphere, which is the heliosphere This is like a bubble within a bubble within a bubble and it takes me to a fractal zone. It's like nature Kind of knows how to replicate a good thing. I kind of want to embed that idea. This is how stars have astrosphere. So everything you look around has this zone of magnetic field that we can see, we can touch. And so my final comment to you is that you live inside a star's gigantic magnetic embrace.

It keeps you safe. Do not forget that. And thank it. Look at that. That's pretty cool. Now, it makes perfect sense that people used to worship the sun. Because they knew. They knew. They knew. They knew more than we did. They were like, hey man, that thing is keeping us safe. All right. We need to show it a little respect. So, so, Leica, catch me up on the Parker Solar Probe. Because last night, I encountered it. I was telling everybody

about how it achieved a speed record for any human-made object. Because there were some trajectories as it fell towards the Sun where the Sun accelerated it. And it was some still very small fraction of the speed of light, but faster than anything had been sent before. But the Parker Solar probe, what was its mission and did it accomplish it? And is it still going? You know, I'm answering the last question first. Yes.

Parker Solar Probe is going strong. The first seven years it launched in 2018, the first seven years were basically dedicated to doing the prime mission and after that NASA is going to decide it as operating mission. So what Parker Solar Probe has done now after 24 orbits, it took it to its closest point called the Perihelion. something like close to 10 solar radii, 3.8 million miles. But I know things in solar radii. It's inside of 10 solar radii. And that was important because we wanted to have a little bit of idea of crossing over from the alpha zone into the corona. where the particles actually become solar wind. So we've done 24 of those and reached the closest point. And now we just kind of finished, I think, 28th per helion in June. Wait, wait, wait. Lika, is that closest point within the corona? Or is it still kind of safe outside the corona? Because you said 8 radii that feels a little distant. Fancy would ask that, Neil. That's such a binary question.

Seriously. Damn. Just getting dissed on my own shelf. So, I already told you right that the Alphen surface, Alphen zone is not a spherically symmetric zone. It's not at 5 solar, radii, 10 solar, radii. We made lots of assumptions. There's not a sign there saying entering the Alpenzone, right? No. So we are actually close enough that we go in and out. And we are actually able to measure the pristine solar wind in the corona. That's very cool. That's the best of all world. That's how it was designed. If you're only in it, you just... Getting the same measurement. Exactly. What good is that? That's really cool. Very cool. Very cool. Instead of going to the poles of the sun, that was the original idea of Solar Probe. It said, hey, we are going to the sun. We are going to take and that would have taken very little data and said, no, we want data. And so we created this alternate concept in the ecliptic and we get Venus gravity de-assisted momentum. So it's between Venus and the closest point to Sun and that point is 3.8 million miles, but let's let's stick to Solar area. Yeah, the Sun's diameter is almost a million miles across. Right. Now, Lika said something just casually. Go ahead. But

I got to stop on it and like give an explanation. All right. Okay. She said What's the word you use? D. What? Accelerate. Oh, no, no, no, no. Gravitational de-assist. De-assist. De-assist. De-assist. Yeah, not de-ass. Okay, I'm sorry. De-assist. De-assist. De-assist. As opposed to... We are so accustomed to hearing by gravitational assist. Assist. Because you have a probe that doesn't have enough energy to get to its destination. Right. You come in behind... You sneak up behind a planet. Right. You fall towards the planet. You gain its orbital speed and you keep coming out the other side. Right. So you can do this Boost your speed, get to wherever you want in the solar system. It is a problem. You want to get to the sun. The sun is not in orbit. We're in orbit, and we're going to launch it from here. We got to somehow eat that orbital speed. We got to get rid of it somehow, so that we can fall in towards the sun. Because we don't want to overshoot it. Correct, correct. So, if you come the other direction into a planet, you get a de-assist. Ah, gotcha. Yeah.

And Venus is sitting there conveniently between us and the sun. And so Venus is doing the breaking for us. Doing the breaking for us. That's really cool. So it seems like, I don't know if it's true or not, that there's been more a rural activity visible at latitudes farther south than usual. Right. And your rower, we know, comes from the solar wind. Right. And the solar wind is stoked, we know, from the solar cycle. Okay. Leica, tell me about the solar cycle. I know it's a big question, but how often do we have you? Okay, the solar cycle and what it does to the solar wind and is the sun burping up more wind lately than historically? Or are we just better informed? Because we have space weather people saying, look north, go north. This weekend, Aurora, check it out. There you go. Is it just our awareness or is it really happening?

And what do some verbs smell like? Is it like pastrami? That has to be the next cosmic query question, and I will answer. Not right now, because I have to think about it. But it's a great question, I think. What will it answer? So solar cycle, I mean, I don't even... We talked about this in some of the previous shows, where to begin. But generally speaking, you know solar that the convection zone that we talked about and the convection cells and all that and the churning of plasma and creating magnetic field with dynamo all of these creates kind of condition for generating it is moving

plasma creates magnetic field that's the key thing to remember the same process of dynamo anywhere but in the case of the sun it has a beat It has a couple of beats actually. Well, I just want to add something there. So it's not just moving plasma that creates manganese field. It's any moving charge. Right. And a plasma is a charge cloud. Right. So you get the manganese field for free because they're free charges moving in the plasma. Right. But turbines and things that generate electricity.

They have charges moving within a magnetic field, and it creates a current. But anyway, so it's a broader fact than just a plasma moving through. Well, no, we have one on Earth, but it's because we have an iron core. Oh, and our core, and it's liquid. But it's liquid. Yes. Otherwise, we'd be dead. It's molten. The Earth cools completely. We lose our magnetic field. And when we're done, right? We're toast at that point. We just have to go underground. Oh. And now we're chuds. Now we're chuds. OK. Please continue, Lika. So actually, Kind of this is something, you know, talking about a little bit more. Whenever you hear the term electromagnetic, we really can't separate electricity and magnetism. They are coupled. They are coupled physical process. One generates the other. One's movement generates the other. And that's what you're saying, Neil. And it depends what's moving and what's stationary.

And depending on that, you either create an electric field or a magnetic field. Yeah, there you go. Oh, that's cool. It's very cool. I mean, that's a great way to think about it. That's very cool. Yeah, that they're part of the same thing always. Yes. All right, great. That's great. So you have this turbulence. So all this is happening. And so we get like a couple of signatures from all of these. And that is roughly 11-year solar cycle, where the solar magnetic cycle peaks where we get these sunspots you know big gigantic freckles on the sun they are dark but they are dark not because they are less intense or energetically less but because the magnetic field is so strong, it inhibits kind of outward propagation of radiation. So you see it dimly in comparison to the rest of the surface. So solar max is a period when you have lots of sunspots. They basically start at mid-latitude, 50 degrees north and south. Then they gradually decline, and then we get into solar minimum.

and in solar minimum there are times when you have no sunspot that doesn't mean the sun is dead there is still sort of a basal magnetic field always and these are things we are understanding better but another interesting thing that happens is and that's why I say two beats right so there is the sort of the 11 year solar cycle magnetic field increases magnetic field decreases but there is a flipping of the magnetic field also. So magnetism has two poles, north and south pole. Take a bar magnet and you will see those polarities if you put iron filings around them. So in the case of the sun, what happens, you know, what was say one polarity in one hemisphere, say northern hemisphere, changes completely.

in 11 years. But in 22 years, it goes back to its original polarity. And Neil, you have to explain this. That's why it's called a one magnetic solar cycle as opposed to one solar cycle. So the upward effect of space weather kind of phenomenon doesn't care quite as much. We haven't at least seen its evidence. More data might lead us to understand it better. But they depend on really the intensity of the magnetic field and how many coronal mass ejections are happening, how many solar storms are happening. They happen in each cycle from maximum to minimum. So the space weather cares less about the polarity of the sun than it does about the intensity of the magnetic field at any given time. That's a fair statement. Well, I want to kind of be very clear about it because polarity does matter at Earth, but it is not the polarity on the sun.

So what the solar polarity is doing is less consequential. But the coronal mass ejections when they come towards Earth and interact with our magnetic field, our magnetosphere, then the polarity that the particles are carrying is very important. You know, in one case, you would actually have a reconnection and particles would penetrate where our magnetosphere is no longer a bubble or a shield. In the other case, they are just going to get deflected. So they repel. So comment, please, on the greater frequency of reported aurora. Because I looked at how peaky the last solar maximum was. It was not oddly high. It was maybe even a little low compared with an average over the last 50 years or so, or 100 years. So is this just better reporting? Because it was like 11 p.m. and I looked to my right, I was driving on Long Island, I looked to my right, and the sky was kind of glowing reddish. And I said, wait a minute, the sunset many hours ago, it

can't be sunset colors, Wait a minute, I'm facing north. What the... So I pulled over, pulled out my phone, my camera phone and took a photo, which is... The phone is more sensitive than your eyes are to different colors. And there it was. Aurora? It was Aurora. And I'd never seen Aurora on Long Island before in New York. And so... Do you remember the time? Which month? I do remember the date. It was October 10th, 2024.

And that year was Solar Max, isn't that correct? Yep. Okay. Cool. So are you friends with the Aurora? Like, yeah, I know that one. Well, I have son's number. I could tell you stories about the son. And so this is, you're asking a very important question because it is a question in my mind. And someday when I am not working professionally, I want to go back. to science and actually kind of try to understand this. And I share whatever I experience and my speculative thoughts on this. I'm not going to go there right now, but it is a confluence of two things. We are all very aware. So I would say, heliophysics figure that I talked about in 2024 and overall, the outreach that NASA has done over the years has really sipped in to the community consciousness. So people are paying more attention. We have better cameras. So we are capturing this, but still that doesn't answer everything. But however, I do end every show saying. Keep looking up. You should. You should. No, no, I do. Every show ends that way. So maybe it's us. Right. Just seeing what we should be seeing because we actually did what we were supposed to do. We're looking up. So that's a little piece of it.

We don't know how big a piece keeps going. So the aurora happens when solar wind particles come and interact, you know, with our atmosphere dislodging, you know, oxygen atoms to, you know, their electrons to higher orbits and we get different colors like green at 100, 200 kilometers, red with higher energy. These are all oxygen. at higher elevation and then you have blue and green for nitrogen so we kind of know all this what exactly is going on but remember one thing that auroras are most visible in the higher southern or northern latitudes around the magnetic axis and which is substantially different from the actual geographic north south axis so May 2024 was actually one of those auroras where basically it is the force G5. NOAA sort of describes storms in scales and G5 is the highest geomagnetic storm scale and so that particular the May 2024 aurora was G5 level and it lit up all the way to Ladakh, India to Florida. I mean, think about being

south. It is incredibly south, right? So if you allow me, I kind of want to describe there are a couple of personal stories related to Aurora that hadn't happened until the solar cycle. Oh, please. Yes. One is May 2024. You know, that happened between May 10th and 12th, 2024. I was in DC. I didn't get to see the Aurora. My friends took pictures, you know, as I would have seen it from our deck in Boulder and sent it to me. About a week earlier, dear friend and colleague of mine, Dr. Jennifer Gannon, a space weather scientist from Boulder, passed away. And I was heartbroken.

And I wrote a short piece on LinkedIn essentially celebrating her life. And to me, it felt like that auroral display was nature celebrating her life. And I wrote a little haiku on that. Yeah. And this was actually picked up by White House. And we named our first solar storm as Ganon Storm. Oh, that's lovely. We meaning, you know, not anybody I know but White House people. Space with our people. This is incredible. You know, we have named hurricanes for 100 years. But this is the first time that we named a storm. Well, that's beautiful. That's beautiful. Well, thanks for sharing that. There's one more. Oh, another one. OK. Well, because they are so personal, I cannot not say it. Please. So in November 2025, now, you know, this is kind of a sad story. You know, in my life, who has looked at the sun, chased eclipses, studied everything, I had never seen an aurora with my own eyes. I had not. In November, between 11th to 13th of November, and this event reached the G4 scale of NOAA. It's like the sun did a private show for me. I saw an aurora so bright from 6.30 p.m. to 3 a.m. in the morning. And our neighbors thought, you know, I had gone crazy, you know, basically.

Now, I myself have been in a position where I've seen colors from 6.30 p.m. to 3 a.m. I'm not sure, I'm not gonna say it was Aurora. It was very beautiful though. The sun didn't speak to you in color, but your mental state did, which is pretty awesome. Okay, that works sometimes too. Why not? Why not? So, the sun has these storms every now and then. They have to be aimed towards Earth for that to affect us in the way we describe for Aurora. Because if it's not aiming towards Earth, then it just misses Earth. Then it goes to Mars or any other place or directly into Heliosphere. I just want to be clear that not every storm observed on the Sun results in Aurora on Earth. So with that in mind, I want to ask both of you I can't remember the name of the movie, but the, I believe the opening scene is these catastrophic

coronal mass ejections that are so massive that now in the movie they say these microwaves come to earth and melt. Like half the damn half a city. And they're like very random. Some things melt, some things don't. That wasn't 2012?

Is that 2012? Well, there's another one where they geologists go to the middle of the earth to find out what happened. That's the core. The core. That's the core. That's the core. Yeah. I think you're talking about 2012. Is that 2012? Yeah, they pulled that out of their ass. Really? Okay. You know what? I kind of, because we know so little, honestly, honestly. So it's not, okay, one thing we can say, we haven't seen this from the sun, we don't expect to see it, probabilistically speaking. But it is not impossible for other stars. We now study exoplanet, right? And I think heliophysics informs the exoplanet science. So we have all these red dwarf stars with gigantic sunspots that we don't see on our sun. And they produce coronal mass ejections and storms that are of the gigantic proportion. And so what they could do? We don't know.

I mean, that's why science is so beautiful. You know, we export our knowledge to any place in the universe. And sun and earth, there are laboratories. Well, we even can bring past data into the present if it can give us insight into what we should fear. For example, I could spend a moment, because we're running a little bit short in time here, but there's stuff I want to make sure we hit. Was it 1859 this famous The Carrington event the famous solar flare that short circuited the brand-new telegraphs that they had along the railroad to communicate from one station to another have we had that event since then and But we were not as susceptible Or would such an event today knock out all of our satellites, which are right now, Elon alone has 12,000 satellites up there? It's actually a fascinating question because one of the things we don't get to measure during such a term, even during Carrington event, I think there was only one magnetometer in Mumbai, that measured the actual magnetic field.

So, something happens on the sun, but something that all those mass, magnetic field particles have to interact with Earth. That is where space weather phenomenon is actually happening, right? Where the ionosphere gets more ionized, where electricity sort of funnels into through the soil into

grids and we get power outages where energetic particles can spoil satellite sensors where astronauts are warned for radiation where high altitude aviation is susceptible again for radiation. What I want to know is it a G5, G10? What level storm will put our satellites at risk? Because we have a lot of assets right now in space. As a scientist in me, Neil, I think we limit ourselves when we put those restrictions. We don't know enough to say because there are many different phenomena interacting with many different environment. There's magnetic field, there's energetic particles, there is radiation. So sometimes even, and I can give you an example. Do you want me to give you an example? Please do. Please. Okay. So.

I think in late January, early February 2022, Starling launched something like some 49 satellites, which it does very periodically. Everybody pays attention to space weather, what the sun is doing and NOAA actually informs everyone. And so they were aware there was a very mild coronal mass ejection going on. So they were familiar and said, no. we can launch into that. They launched and you know about 38 to 39 of those satellites didn't reach launch orbit with this really medium class coronal mass ejection. So what happened? What happened is that the Sun launched many coronal mass ejections that gradually started hitting the thermosphere, ionosphere and created a lot of drag And the Starling satellites didn't have enough propulsion in Delta V to launch them. Look at that. That's yet another way you can impact the health and safety of a satellite. Right. I forgot about that. Oh my gosh. So do we know what level the Carrington event was from 1859? From every indication it was absolutely G5. And I don't think we have the scales right.

we get saturated, so we need to really think deeply. And I think NOAA and space weather communities thinking about it to really amplify the scale. Yeah. Plus the, there are a lot of military satellites that are radiation hardened. So there are ways you can sort of and there are others. For example, you can, we have some of our telescopes, depending on whether we're going into a a meteor shower or some other phenomenon, we can actually put the telescope in safe mode. We would close it down, turn it over, and then so there are ways to mitigate this. That's why prediction is so important. Space weather. People just live in space weather pages. People don't talk about space, whether the way we talk about our weather, hot

enough for you? No, they don't do that. So it's that real consequence. That is going to change, Chuck. Mark my word. Get me back in ten years if I'm still here and not blended with the corona. And I'll prove myself right. If you blend it with the corona, it would become that much more colorful. Oh, look at that. Oh, Neil.

Thank you. That was good. That was good. That was pretty good. All right. What's coming down the pike in the future in missions to the sun? So the community, the world has launched, you know, lots of interesting missions. Punch, polarimeter to unify the corona and heliosphere. It's a constellation of four small satellites in polar, sun synchronous, low Earth. orbit and they image the solar wind. That's a gap area in imaging, in understanding how solar storms are born on the surface of the Sun and propagate through the corona into the planetary medium and touch us here at Earth. And that is really going to connect in imaging from the surface of the Sun all the way to Earth and it's extraordinary science. You know, we can connect it to auroras in future. Then we have things like IMAP, right? Interstellar Mapping and Acceleration Probe and that is actually looking at the heliosphere. The original question, it operates from Lagrange Point 1 with instruments and it's kind of, it's still, it started taking science data I think in February.

And it is trying to sharpen the discovery that one of the earlier missions called IBEX, Interstellar Boundary Explorer found, which was discovery. All of a sudden, you look at the heliosphere from inside the heliosphere, inside the bubble, and you see this giant ring in high-energy, energetic, neutral atoms. So IMAP is going to shed light on that, but it also serves as a space weather buoy. I see what you did there. You said it was shed light on it. Yeah, I see that you catch that And so what you implied there is these missions that they can't use visible light because not all of this is giving us the imageable Substance invisible you need other bands of the spectrum to image the high energy, right?

What bands of light ultraviolet x-rays I guess well the ultraviolet x-rays white light infrared all of those outside of that Yeah, I mean some of these you have to be outside of Earth's atmosphere. Otherwise they get absorbed. So that's why we have to launch You know before we launch sort of

extreme ultraviolet and x-rays satellites. This is way back, right? We did not really know that the sun was a cauldron, as you described very early on, of this sort of seething plasma. You can't see that in white light. So, Leica, is the United States a leader in the world in studying the sun? Or what other countries are players here in this space? Sun belongs to no one. And sun shines on every flag. There are very many interesting missions from all over the world. Europe plays a very big role. European Space Agency, Solar Orbiter, which is in partnership with NASA. And it takes us up to 30-35 degrees above the ecliptic so we can view the poles of the sun. Never before done before.

We have proba 3 also in Europe that created two spacecraft to create artificial eclipse externally occulted coronagraph at extreme with two spacecraft And they are finding out again sort of new sources of slow solar beam The thing is every time you look somewhere you find something. So this is eclipses on demand I love that eclipses on demand yeah, uh India launched Aditya L1. Aditya is sun in Sanskrit, which is parked at L1, looking at the surface of the sun, interior with magnetic field of the sun, with in situ instruments. So Aditya is Indian's eye on the sun. We have JAXA. JAXA, the Japanese space agency. And then there is China. China is doing amazing stuff. So Kua Fu L1 mission that was conceived of about 20 years ago. It's launch. It's measuring all kinds of phenomenon, including X-rays. So what every agency is doing is looking at the gap.

areas that, you know, a while ago we had as a community created and trying to fill those gaps. Is this a cooperative effort? Is this information shared around the world with all the other scientists? So, except for China, everything else is shared. Except for China. Right now. Right now. And by the way, the Artemis Accords, which are a modern day attempt to create some level of cooperation in space. In particular, if you discover something that could impact the safety or security of my next mission, either my hardware or my wetware, people, the brains, then you are obliged to share that. But China, last I checked, had not yet signed it. But India has signed it, and the list of countries you'd expect have signed it. And the list of countries you might expect wouldn't have signed it yet.

That'd be Russia, Russia, yeah. Yeah, of course. Sunlight is spherically symmetric, shines everywhere, right? Everywhere. That's right. But how it touches us as human being, our culture, it was, that's like that heliosphere, heliopause border, or the alphanic zone border, it's very crooked. There you go. Ooh. Look at that. That was profound. That's a mic drop. That's a mic drop. We like it. Well, Liga, that's all the time we have. This has been a delight to catch up with you. And thanks for sharing those two stories. Sad but beautiful. And about your friend and about yourself. Yeah. And what role Aurora played within it. We'll have to catch up with you again. Just, you know, if the sun burps or anything or misbehaves, you're going to be on our hotline, okay? Well, when you get those cosmic query questions, you know.

Oh, there it is. Absolutely. On the sun, heliosphere, because everything is under that. There you go. Everything under the sun. Under the sun. All right. Chuck, always good to have you, man. Always a pleasure. All right. This has been Star Talk. This was the heliophysics edition. Nice. I like that. Until next time, I bid you to keep hooking up.